

Brains Are Not Computers

Celebrating the Achievements
of Robert Kozma



by Bernard Baars,
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In this article, we provide an overview of Dr. Robert Kozma's accomplishments, based on essays by leading experts in various areas related to his research, covering mathematical, computational, and cognitive aspects of his neurodynamics modeling and applications in biologically inspired autonomous systems and robotics. Specific contributions are given by Dr. Miklós Ruszinkó (the Rényi Institute of Mathematics, Hungarian Academy of Sciences) on mathematical and graph theoretical aspects of brain modeling, detailing Dr. Kozma's work, which is now known as *neuropercolation*. This is followed by recollections from Dr. Roman Ilin [U.S. Air Force Research Laboratory, Wright Patterson Air Force Base (WPAFB)], a former student of Dr. Kozma's, on computational models of spatiotemporal neurodynamics, and the work Dr. Kozma has made jointly with the late, world-renowned Prof. Walter J. Freeman [University of California (UC), Berkeley] and his students in a field now called *Freeman K Katchalsky-sets*. Dr. Edward Tunstel [the United Technologies Robotics Center (UTRC), East Hartford, Connecticut] provides an overview of biologically inspired robotics and several unique applications Dr. Kozma has made in this field together with colleagues from NASA Robotics. Dr. Bernard Baars (Society of Brain-Mind Sciences, La Jolla, California) contributes a cognitive scientist's perspective on Dr. Kozma's work with an emphasis on implications of understanding and modeling consciousness. We believe this multiperspective

overview provides an insight into the truly interdisciplinary character of Dr. Kozma's remarkable research achievements.

Dr. Kozma is a professor of mathematics and the director of the Center of Large-Scale Integration and Optimization Networks (CLION), FedEx Institute of Technology, the University of Memphis, Tennessee. The research conducted at CLION is focused on develop-

ing advanced optimization techniques based on biologically motivated and cognitive principles in large-scale networks. He also holds the position of visiting professor at the School of Information and Computer Sciences, the University of Massachusetts, Amherst, where he currently leads the new Defense Advanced Research Projects Agency (DARPA) initiative toward superior artificial intelligence (AI) at the Biologically Inspired Neural and Dynamical Systems Laboratory.

Dr. Kozma is a Fellow of the IEEE and the International Neural Network Society (INNS). He is a recipient of the INNS Gabor Award (2011) and has been a National Research Council senior fellow (2006–2008). His research has been supported by various agencies and corporations, including the DARPA Physical Intelligence and Superior AI programs; the National Science Foundation Collaborative Research on Computational Neuroscience Program, and Biological Information Technology Program; NASA's Revolutionary Computing Program; the Air Force Office of Scientific Research Mathematics and Cognition Program; the Air Force Research Laboratory Sensors Directorate; as well as the NASA Jet Propulsion Laboratory (JPL), the National Research Council, the FedEx Corporation, and the First Tennessee Foundation.

He has more than 35 years of research experience in the design, analysis, and control of intelligent systems. His research focuses on developing novel artificial intelligent approaches motivated by the operation of brains, and his interdisciplinary approach comprises applied mathematics, statistical physics, computer science and engineering, neurobiology, and cognitive science. He developed fundamental mathematical theories of evolving random graphs and networks indicating spatiotemporal discontinuities in the dynamics of these systems. He leads broad, interdisciplinary efforts aimed at the implementation of these breakthrough mathematical and computational principles for the development of robust decision-support systems in a wide range of application areas, including sensor networks, autonomous control and decision-support systems, large-scale data analysis and knowledge extraction, brain networks, brain imaging, and brain–computer interfaces.

Here, his main accomplishments are summarized.

- ◆ In the area of brain modeling and data analysis, with his interdisciplinary team of mathematicians and

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neuroscientists, he established the neuopercolation model, motivated by the multilayer structure and dynamical properties of the cortex. The key innovation of neuopercolation is its introduction of criticality to describe brain behavior and brain structures. This model operates through phase transitions between dynamic brain states, including background

activity, narrow-band and broad-band (chaotic) oscillations. Neuopercolation can be used to model and potentially predict rare events (catastrophes) with sudden, drastic changes in the behavior of complex systems, such as stock market crashes, solar flares, earthquakes, and other disasters. Important implications of the corresponding theory concern the prediction of the rapid switch from one state to the next, described as *dragon kings*, as opposed to the unpredictable nature of *black swans*, which represent self-organized criticality with self-similar dynamics.

- ◆ In experimental brain research, he worked closely with Walter J. Freeman for more than 20 years, until Dr. Freeman's death in 2016. Their joint research characterized the intermittent singular space–time dynamics observed in brain-imaging experiments, which demonstrated transitions from metastable, high-dimensional background states with large-scale synchrony, to brief episodes of low-dimensional states with the collapse of synchrony. Neuopercolation helps to interpret the collapse of the high-dimensional cognitive space to a low-dimensional subspace of attractors. Such a collapse demarcates the onset of awareness and conscious experience, the “aha” moment, as part of the intentional action-perception cycle. Applications include novel brain–computer interfaces, in which transient synchronization dynamics are employed to identify the formation of percepts and to quantify high-level cognitive experiences. Analyzing electrocardiogram/electroencephalogram (EEG) data, they identified synchronization effects with sudden transitions in spatiotemporal neurodynamics, which are interpreted as cognitive phase transitions. Novel scalp-EEG designs are proposed with high-density electrode arrays, which provide the hardware and software tools necessary to drastically reduce the effect of muscle artifacts.

Dr. Kozma's results indicate that brains are not computers in the sense of our presently dominant digital computer paradigm. He proposes a novel computation paradigm, which goes beyond symbol manipulation by a given rule set, as specified by the Turing principles. His model operates sequentially, resembling Turing machines, however, it has several distinguishable features that surpass

Turing computing: 1) the computing sequences are self-organized through frequent phase transitions between synchronized and desynchronized states; 2) the symbols are not fixed but dynamically emerge in the form of spatially distributed, metastable patterns; 3) there is no pre-defined instruction set advising how to switch from one pattern to the other, rather, the system follows a dynamic trajectory in a high-dimensional state space, modified by learning experience. Some of his theoretical results have been used to develop new intelligent algorithms for robot navigation and control. In collaboration with colleagues at the JPL and under the leadership of Homayoun Seraji, he developed a novel autonomous control for the Mars rover prototype SRR-2K. Applications also include distributed sensor systems in the framework of the AFRL-layered sensing paradigm.

Brains Are Believed to Be Critical, Miklós Ruzinkó, the Rényi Institute of Mathematics, Hungarian Academy of Sciences, Budapest

Prelude

"Brains are believed to be critical," said Robert, sipping a beer during one of our late-night discussions in Memphis, Tennessee.

I first met Robert in 2011, while visiting Béla Bollobás, one of the leading mathematicians at Cambridge University and the University of Memphis. We had several interesting discussions, during which he explained to me his novel results in brain research while I thought about related mathematical theories—a real brainstorming. Unfortunately, due to serious illness, I left Memphis earlier than expected, which caused a break in our discussions.

Thoughts About the Brain and the Szemerédi Lemma

Nearly two years later, I again visited Béla Bollobás and we continued our intriguing discussions with Robert. At that

time, I was working with Endre Szemerédi to solve several long-standing open questions using a strong mathematical tool, the famous Szemerédi lemma [2]–[6]. In 2012, Endre Szemerédi won the most prestigious prize in mathematics, the Abel Prize (considered the Nobel Prize in mathematics). This famous lemma [9] is often cited as "there is an order in every chaos." Indeed, if we look at a structure from the "correct angle," we will notice the order. A bit more precisely, it discovers a small "skeleton" in every large graph, which is already responsible for the main properties of the graph. This skeleton can be used for obtaining the desired structural information of the graph; this information can then be transformed to important information in the original graph. Essentially, it says that every large graph can be partitioned in a small number of parts such that nearly all of the pairs behave like a "random bipartite graph," i.e., the edges between the regular pairs are spread quite evenly. This phenomenon is depicted in Figure 1. If we look at the unstructured mess in Figure 1(a) from the right angle, it can be seen as rather structured in Figure 1(b), and the edges between the clusters form the "skeleton" in Figure 1(c), describing the major properties of the original "mess."

Clearly, such partitions simplify our image of the graph. In other words, one can see the important properties of the graph and avoid the making the mistake of "not seeing the forest for the trees."

The message of this central and relatively new result in mathematics for a brain researcher is clear. Robert immediately realized the importance of seeing the skeleton of the billions of neurons with their interconnections. Indeed, one could derive important information from analyzing a relatively small skeleton that possibly contains only hundreds or thousands of clusters. The main problem is that this lemma works for huge structures only, far beyond billions of nodes, e.g., Gowers in [1]. This appears to put limits on the practical usefulness of this method, so why do we still think that it can be useful in more practical applications, too?

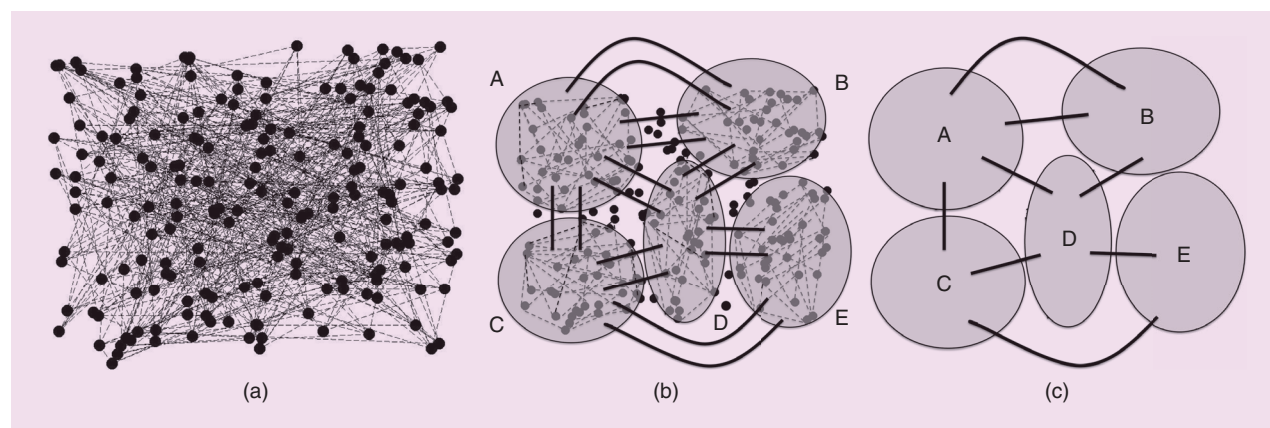


Figure 1. An illustration of the Szemerédi lemma [10]. (a) The graph displayed here looks like an unstructured mess and is the original graph. (b) Looking at the same graph, structure can be discovered in the form of pairs of vertices in a few clusters. (c) The edges between the clusters form the "skeleton," describing the major properties of the original "mess."

The following reasons provide the answer.

- ◆ Gowers' complicated example does not mean that for those graphs (networks) that occur in real life, the lemma does not work if the number of vertices is not astronomically large.
- ◆ It is known that every graph can be well-approximated by graphs that have Szemerédi partitions with not too many parts.

Moreover, we turn to our advantage the phenomena that, in certain cases, an astronomically large number of vertices is required to satisfy the lemma. In brains, small perturbations in its numbers of connections may cause the original highly connected brain network to break apart into several structural or functional components. This suggests that before the changes, the Szemerédi lemma is applicable, i.e., there is the highly structured skeleton. On the other hand, after the changes, the Szemerédi lemma no longer applies. This may indicate some structural property of the original graph in this indirect way as follows:

- ◆ determine the class of graphs for which the Szemerédi partition already works, assuming that the number of vertices is not astronomically large
- ◆ approximate the general graphs with those, which have the Szemerédi partition with a reasonable number of parts
- ◆ obtain the number of perturbations (i.e., edge additions and deletions) needed to turn a general graph into a one, with few parts in a Szemerédi partition and vice versa
- ◆ determine the phases in which the brain network is either structured or unstructured.

This ongoing project is ambitious, very important, and shows the relevance of interdisciplinary discussions and Robert's delicate instincts in this field.

Brain and Dynamics

"Brains are believed to be critical." It took me a while to understand the meaning of this sentence. In terms of mathematics, this basically means that a tiny change in the structure can completely change the outcome of a process over this structure; this is usually called a *phase transition*. A typical example with a double jump is the emergence of the giant component in the Erdős–Rényi random graph model.

Following several long discussions, we invented the $G_{Z_N^2, p_d}$ random graph model—which turned out to be an interesting methodology in brain research [7]—to represent the function of the brain [8]. Graph theory has been successfully employed in brain network models, however, the previous studies did not include the main property of connections, i.e., the distance dependence of graph edges to model long-range axonal connections. We introduced a random graph model, which has a stated property, i.e., where the probabilities of edges are defined with respect to the distance of vertices to be joined. On the constructed graph, we considered the propagation of activity in

populations. The precise results obtained show the presence of parameter regions with convergent behavior (fixed point), as well as multiple fixed points (both stable and unstable), indicating oscillations. $G_{Z_N^2, p_d}$ turned out to have very useful properties itself, e.g., a linear expected number of added random edges indicating an easy control over the proportion of long-range and short-range axons, Poisson degree distribution, and a logarithmic small-world property, verifying that signals in the brain spread very fast through long and short connections in a hub structure.

The criticality of the process defined on this graph is completely understood in the real model and its mean-field approximation. Our approach is completely novel in several aspects, and its most important feature may be that the process is not monotone, which is technically difficult to handle, because the activation cannot be considered vertex by vertex, i.e., only in full rounds.

Concluding Comments on Mathematics and Interdisciplinary Research

Being a mathematician, I tend to be skeptical about some parts of the interdisciplinary research where trivial mathematics is used to make a paper look more sophisticated than meaningful. In our work with Robert Kozma, we attempted to use hardcore mathematics to obtain predictive models. I am very grateful for his inspiration and I hope that we will be able to continue working together on this line of investigations.

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Modeling Neural Population Dynamics, Roman Ilin, U.S. Air Force Research Laboratory, WPAFB, Ohio

Neurodynamics and K-Models

Artificial neural networks (ANNs) have permanently taken the spotlight from the field of AI, using the name *deep learning*. Most people know that ANNs are inspired by the biological brain, but few nonexperts understand that their architecture is far removed from biology. Robert Kozma spent considerable time developing biologically plausible brain models that reflect observed brain dynamics with a practical application. The starting point of his work was research conducted by Walter J. Freeman, a neuroscientist who pioneered multielectrode recordings of EEGs in small animals. Freeman observed that sensory inputs, such as smells, could not be traced to specific activations of neurons; rather, they created spatial-temporal patterns of EEG activity that are correlated with the inputs. This led to a theory emphasizing neuronal group dynamics as the key to understanding the brain.

Mathematically, the brain dynamics observed by Freeman were modeled after a hierarchy of models called *k-sets*, named after Aaron Katchalsky, a pioneer of neurodynamic research. The models are named K0, K1, ..., K4 [1]–[4]. K0 models are a homogeneous population of neurons, and the subsequent models are networks of interacting K0 models. There are two types of measures of neural activity: average pulse density and average wave amplitude. They are related by a population transfer function, which is determined by using experimental measurements. The dynamics of the K0 set is then given by the following differential equation.

$$\frac{d^2 y_i(t)}{dt^2} + \alpha \frac{\partial y_i(t)}{\partial t} + \beta y_i(t) = \beta \sum_j w_{ij} Q(y_j(t)) + P_i(t). \quad (1)$$

This equation establishes that the dynamics of one K0 population indexed i , y_i is the pulse density of the i th population of neurons (referred to as *activation*), Q is the population transfer function, P_i is the external input, and w_{ij} are the connection weights between other K0 models and this one. α and β are constants.

Figure 2 illustrates various aspects of k-sets [5]. Figure 2(a) shows the shape of the transfer function Q .

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Multiple K0 sets coupled through the Q function generate complex patterns of oscillations similar to what is observed in the EEG. Coupling K0 sets with excitatory (positive) weights creates nonzero equilibrium; these models are called *K1 sets*. Periodic oscillations are generated by coupling K0 sets with both excitatory (positive) and inhibitory (negative) connections. These are called *K2 sets*, as shown in Figure 2(b). The studies of K2 sets by Freeman and later, Kozma and Ilin, demonstrated the existence of multiple, stable limit cycles with characteristic frequencies dependent upon the connection weights [6]–[9]. Coupling multiple K2 sets with both excitatory and inhibitory connections and introducing biologically plausible time delay, results in more complex oscillations with periodic and even chaotic dynamics. These K3 sets model the dynamics of a large portion of the brain, such as the olfactory system, which is responsible for smell recognition.

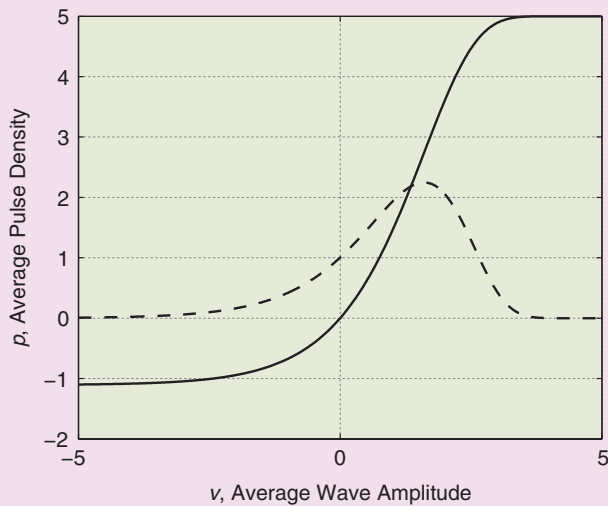
K-Set Modeling

K-set research aims to answer several theoretical questions. Parameter selection is an important step in generating brain-like dynamics, with initial works utilizing parameters estimated from biological considerations. Subsequent research focused on the systematic understanding of k-set dynamics by means of bifurcation analysis, both theoretical and numeric. Studies identified parameter regions that are suitable for simulating near-chaotic oscillations present in the brain, as shown in Figure 2. The nonlinearity of all building blocks makes the dynamics not fully predictable. The return plot indicates presence of a strange attractor resembling a Rossler system (see Figure 3).

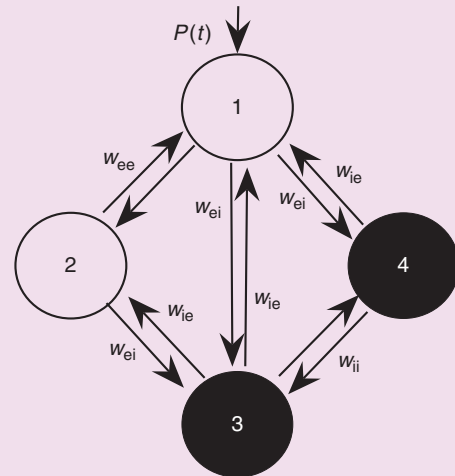
Dr. Kozma studied the role of noise in the creation and sustainment of chaotic dynamics. Simulated dynamics tends to converge with a periodic attractor, and one of the research objectives was to confirm the hypothesis that noise plays a crucial role in maintaining the system in the state of wandering between various attractors, behavior termed as *itinerant*.

The human brain operates in a cycle where the reception and processing of sensor inputs is followed by an interval of passive basal activity when the external input is not received. During the passive interval, the neurodynamic approach proposes the existence of chaotic activity. Once this input is applied, the activity converges with a periodic attractor corresponding to the meaning of the sensory input. This periodic attractor is observed in the EEG patterns in Freeman's experiments.

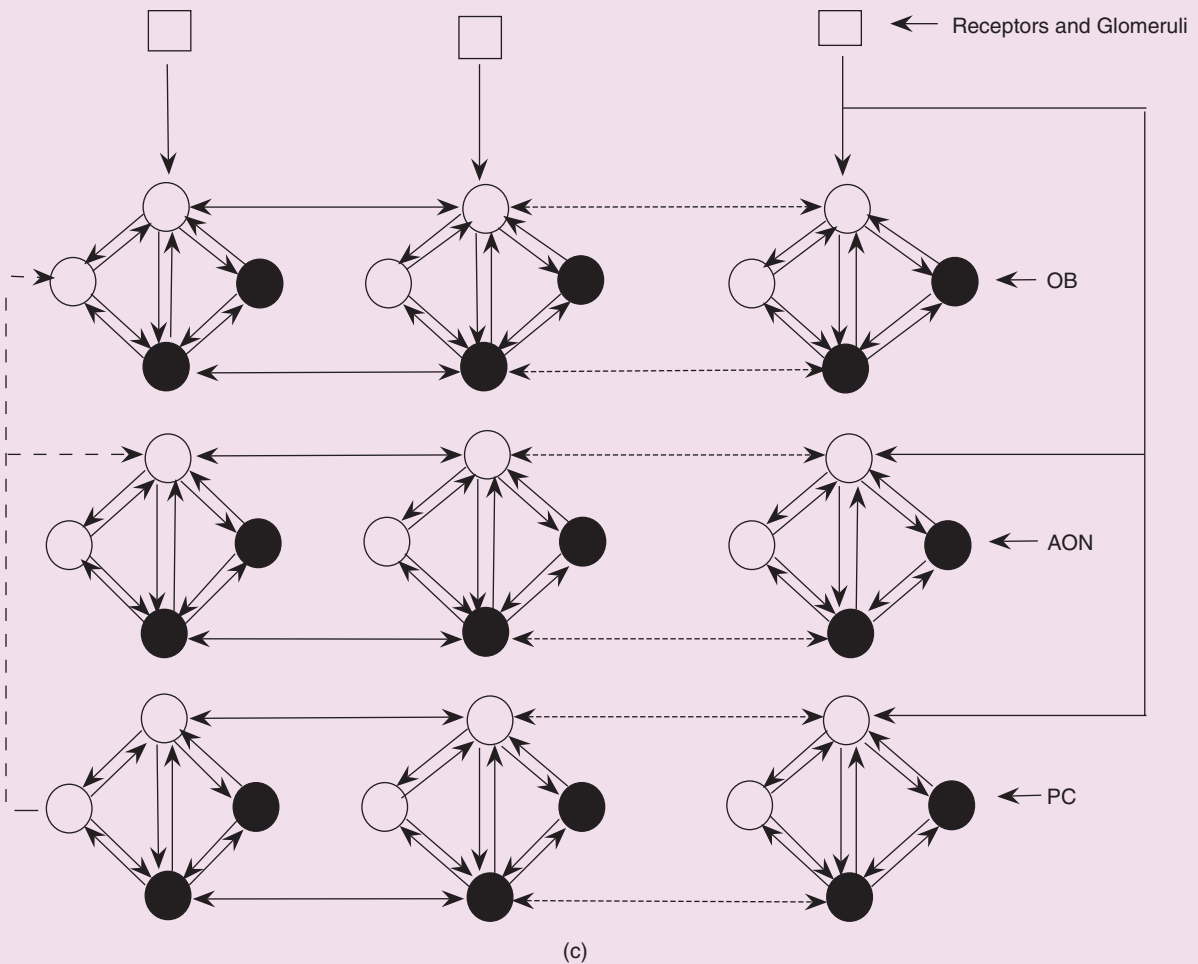
Work in Robert Kozma's lab focused on implementing a pattern-recognition system based on the k-set



(a)



(b)



(c)

Figure 2. An illustration of the k model. (a) The nonlinear transfer function (solid line) and its derivative (dashed line). The nonlinearity of the transfer function is the key to modeling brain dynamics. (b) An example of a K2 set, nodes 1 and 2 are excitatory (white circles), and nodes 3 and 4 are inhibitory (black). (c) The K3 system and its relationship with the anatomy of the olfactory system. OB: olfactory bulb; AON: anterior olfactory nucleus; PC: prepyriform cortex. Dashed lines show delayed feedback; adopted from [5].

dynamics. This system implemented two learning mechanisms that are conjectured to operate in biological systems. Hebbian learning is a mechanism that strengthens connections between simultaneously firing neurons, while habituation learning is the process of slowly weakening the neural connections. These processes involve their own tunable parameters. The implementation of such learning systems created opportunities for practical applications.

Applications to Robot Autonomy

Learning in k-models was applied to robot navigation. The k-model architecture was extended to the K4 model, which is a combination of several K3 models. This architecture models several brain parts used in planning. Sony Aibo robots were fitted with the K4 brain to learn how to avoid obstacles and find targets in a simple environment that was set up at the University of Memphis. The robots learned to select appropriate actions by combining sensor signals from vision and touch with the reinforcement signals [10].

The K4 architecture was also applied to an experimental version of the NASA Mars rover at the JPL. In this case, the sensors included a video camera and accelerometers. The task of the robot was to traverse rough terrain in a specific direction, avoiding obstacles like big rocks that would cause it to stop or smaller rocks that would cause a strong vibration. The K4 model was successfully incorporated into the rover's operating code. Robert Kozma conducted multiple experiments demonstrating the ability of the K4 to learn appropriate navigation actions based on sensor and reinforcement inputs.

Discrete-Time Models

The K0 model is governed by a simple differential equation based on experimental measurements. This dynamic is the result of interactions among thousands of neurons. One of the research directions in Dr. Kozma's lab was the lower-level modeling of brain dynamics as probabilistic cellular automata. This approach was termed *neuropercolation* [4] and was based on the idea that brain tissue is uniquely structured with a mixture of short-range and long-range connections between neurons. This is modeled as a lattice of probabilistically firing neurons with a certain fraction of long-range connections. Large-scale computational experiments were conducted to understand the dynamical properties of such lattices as well as their relationship with the k-models. Neuropercolation models give rise to macroscopic oscillatory activity similar to the dynamics of K1 and K2 sets.

Conclusion

Studies of brain dynamics modeling by Dr. Kozma and his students spanned more than a decade and resulted in numerous workshops and publications, master's and doctorate degrees, and most importantly, advancing the level

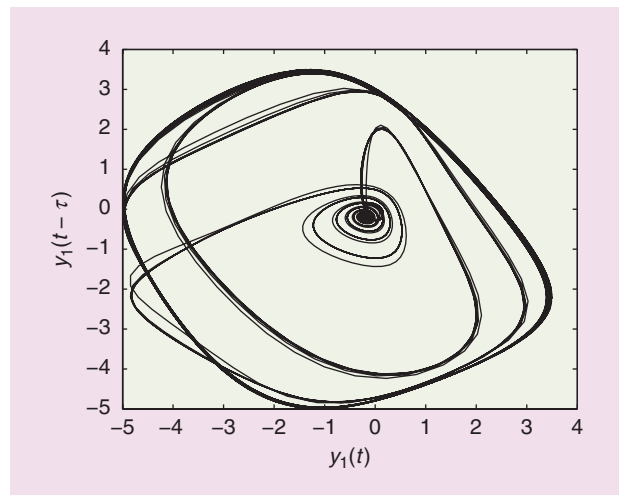


Figure 3. The return plot of activations for one of the nodes of a K3 system in [5, Fig. 2(c)]

of understanding of the brain's biology. These studies represent a unique combination of biologically plausible brain modeling and applied computer science. Beginning with Freeman's ideas, which attributed importance to what he referred to as *mesoscopic dynamics to cognition*, Dr. Kozma's research expanded into multiple directions, producing fruitful results and offering alternatives to conventional brain modeling.

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Robotics—Behavior to Brain Inspiration, Edward Tunstel, UTRC, East Hartford, Connecticut

Biology's Influence on Robotics

Biological inspiration has had a rich influence on robotics. This comes as no surprise because the frequent goal of robotics research and applications has been to achieve some semblance of insect or animal behavior in robots in a relatively simple or otherwise efficient manner. Such semblances are often sufficient for achieving effective motion control or navigation functionality toward the goals of reaching desired locations in an environment and/or for exhibiting a capacity for resilience or survivability in unfamiliar environments. Another commonly pursued goal has been to achieve human-like reasoning in robots and even intuitive or social behavior when interacting with humans.

As early as the 1940s through the 1960s, researchers used various means to instill insect and simple animal-level intelligent behavior in robots. Early examples were Norbert Wiener's phototropic robot Palomilla, Grey Walter's cybernetic tortoises, and mobile automata developed at Johns Hopkins University's Applied Physics Laboratory that included analog artificial neuronal circuits based on sea slug neural-firing patterns. As sensing, actuation, control, and intelligence technology advances, the sophistication of robotic systems continues to increase, particularly when leveraging bioinspiration. Bioinspired robotics overlaps with the IEEE SMCS field of interest in several respects, specifically as it pertains to the integration of theories of communication, control, and cybernetics and their applications to the analysis and design of biological systems at both the hardware and software levels. More recently, many members of the SMCS have been engaged in various aspects of bioinspired robotics. In particular, Dr. Kozma and his students have influenced the advancement of bioinspired robotics with contributions that have derived inspiration from spatiotemporal neurodynamics as well as the emergence of intelligent behavior used for autonomous robot control.

Bioinspired Robotics

Biologically inspired robotics is a broad subfield of robotics research that leverages studies of biological systems to enable and improve the design and realization of robots. Bioinspired robotics has also focused on using robots designed to mimic biological characteristics for the purpose of facilitating our increased understanding of biological systems. In some cases, robots are designed to mimic

biological systems without any particular inspiration derived from deeper study. Bioinspiration goes beyond biomimicry to leverage concepts learned from natural biological systems and applies them to realize the desired functionality of typically simpler robotic systems.

With regard to robotic systems, bioinspiration comes in multiple forms influenced by the manner in which biological systems appear in physical form, the way they locomote or behave as well as what can be learned from their neural and brain functions. Biological system morphology can influence the design of robots intended to operate in environments to which the inspiring physical form is well-adapted. This produces an efficient design forged by nature. Robots can benefit

by mimicking physical form and/or taking inspiration from the motion that that form enables. Examples include undulated motion for robots with slender bodies, fins or wings for robots in fluid media, and multijointed limbs for robots on land.

The particular aspects of biological locomotion, whether considered along with morphology or not, have inspired the manner in which robots move about. The various actuation or control mechanisms governing how biological systems move on land, in air, on and under water, and so forth, have inspired many robot locomotion designs

and automatic control solutions. Sources of inspiration have ranged from biomechanics and musculoskeletal aspects to observations of the maneuverability achieved.

The varied behavior of biological systems that enables them to manage robust existences in complex environments has been a rich source of inspiration. Behaviors observed in nature have inspired the computing architectures and computational algorithms that foster the exhibition of artificial behavior in robots ranging from stimulus response or reflexive/reactive behaviors to more sophisticated intelligent behaviors that respond to perceived situations or context. Beyond observing biological systems, bioinspiration has also come from the scientific understanding of neural and brain functions, leading to realizations of a similar range of robot behaviors and capacities for learning. Robot learning, in this context, is supported by architectures and algorithms that emulate structures or computations associated with networks of neurons and/or that of the various regions of biological brains. It is the latter two forms of bioinspiration, i.e., behavioral and neural, that exemplify Dr. Kozma's robotics contributions.

Brain-Behavioral Intersection

In the 1980s, the behavior-based paradigm of real-time AI provided an ideological framework for efficiently implementing bioinspired robot-control systems. Canonically,

Biological system morphology can influence the design of robots intended to operate in environments to which the inspiring physical form is well-adapted.

behaviors were simple, computational mappings from sensor inputs to motor controls, with each behavior executed to actuate the observed robotic motions to exhibit a specific behavior (e.g., follow light, avoid obstacles, seek sound, wander, and so on). The seminal subsumption architecture introduced by the Massachusetts Institute of Technology was structured as a cascading set of sensor-driven behavioral layers in which the activation of one behavior would subsume the behaviors in other layers based on their relative importance. The ideas about hierarchical organizations of behavior and intelligence in animals provided further inspiration for intelligent robot controllers. The flexibility of the paradigm allowed for fairly open, computational implementations of individual behaviors such as input-output (sense-act) mappings ranging from analytical expressions to knowledge- or rule-based constructs. The behavior representations employing the production-rule (if-then) approach of AI expert systems was not uncommon; this led to a natural consideration of rule-based expert systems based on fuzzy logic as a means to both capture expert knowledge governing intelligent behavior and to accommodate uncertainty in the knowledge as well as in robot sensing and/or actuation.

Taking inspiration from ethology in the form of hierarchical model decompositions of animal behavior led to effective implementations of mobile robot controllers for indoor and outdoor navigation [1]. This research revealed approaches for augmenting the hierarchy with weighted decision-making among multiple behaviors, allowing for simultaneous and partial activation of behaviors and moving from single action/behavior selection to intelligent coordination of multiple behaviors that are simultaneously active to varying degrees. Conceptually related work was introduced as the Biologically Inspired System for Map-Based Autonomous Rover Control (BISMARC) [2] and was motivated by the advantages of hierarchical action-selection mechanisms over purely reactive behavior-based solutions. BISMARC represented behaviors for robot motion, health, and resource management within a free-flow hierarchy with fixed (later, also adaptive) weights governing hierarchical action/behavior selection priorities.

Such hierarchical fuzzy behavior-based systems exhibited characteristics of nonlinear activation dynamics [3] that were similar to phenomena observed by neurobiologists who measured animal brain activity during transitions between distinct behaviors (e.g., in rabbits, cats, monkeys, and birds). This parallel between patterns of behavioral activation signals from artificial, computationally intelligent behavior in robots

and those of biological neural brain signals brought bioinspiration full circle in a sense.

The observations piqued Dr. Kozma's interest and motivated him to seek collaboration on applying his neuro-/brain-inspired ideas to mobile robot-control applications. The hope was that combining biologically inspired control architectures for robot navigation with brain-like architectures built upon models of brain activity associated with spatial navigation and orientation would enrich future implementations.

Kozma's Bioinspired Robotics Contributions

K-sets and k-models, developed by Dr. Kozma's close colleague, the late Prof. Walter Freeman, are biologically motivated neural structures of complex dynamics used for modeling aperiodic oscillations in brains. They differ from traditional neural networks in that they use nonconvergent dynamics for encoding information rather than operating on the principles of convergent dynamics. The KIV model is the fourth of several hierarchical levels of k-sets intended to have the functionality of planning and action selection useful for goal-directed behavior. We first collaborated to apply Dr. Kozma's research to the problem of autonomously navigating the natural terrain relevant to the Mars rovers (Figure 4). We demonstrated in [4] that the dynamic KIV architecture can be used for robust multisensory fusing with fast learning of goal-oriented behavior and navigation in a simulated Martian environment, without a priori information.

We further applied the KIV model to agent-design-based autonomous operation of intentional agent action, driven by its interaction with the environment [5]. In that work, the focus was on investigating the functionality of the brain's

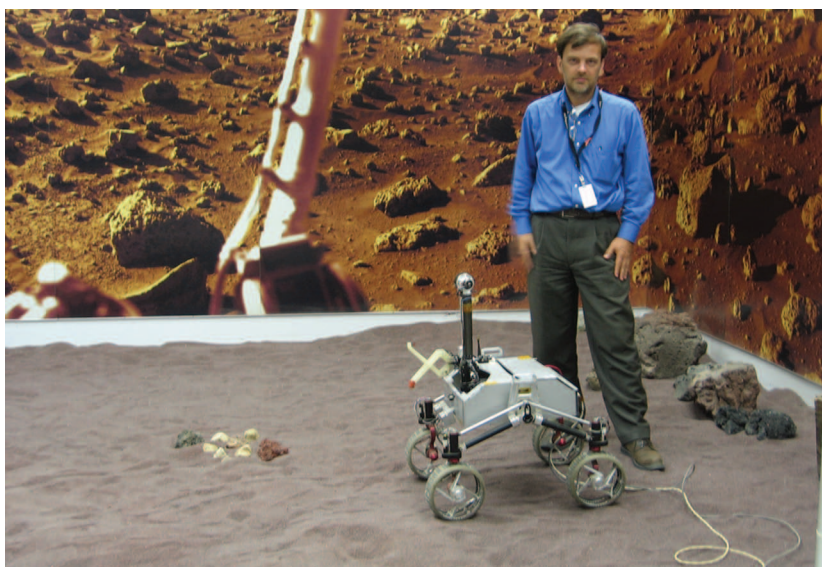


Figure 4. Dr. Kozma with the Mars rover prototype SRR-2K at NASA's JPL in the mid-2000s. His work on biologically motivated autonomous navigation and control on SRR-2K platforms enabled the successful implementation of embodied cognition beyond rule-based artificial intelligence principles.

amygdala and its interaction with cortical and hippocampal parts of the KIV brain model. The results demonstrated that with the addition of an amygdala, a multisensory agent with long-range and short-range sensing is capable of navigating a challenging environment using a KIV-based controller.

We next developed an approach for sensory fusion and decision-making in a network of interacting autonomous agents with KIV as the underlying biological model, taking inspiration from models of multisensory percept formation in vertebrates [6]. Multiagent cooperation among networked robots was achieved in which autonomous units weakly interacted to produce coherent, goal-oriented behavior at the overall network level.

While spending time as a visiting faculty member at NASA's JPL, Dr. Kozma worked with fellow colleagues and myself on his latest robot-control architecture, which was based on principles for implementing a hierarchy of dynamical neural modules with associated decision-making and intentional control. We combined ideas from the hierarchical behavior-based architectures being used at the JPL for planetary rover-navigation technology-development projects with brain-inspired approaches based on k-sets within a single, integrated, biologically motivated framework. By then, Dr. Kozma coined the KIV-based architecture known as *Self-Organized Ontogenetic Development of Autonomous Systems* (SODAS). With SODAS, he brought the means by which to model brain organization and learning. In [7], we developed mechanisms for learning the hierarchy's weights aboard the rover's processor, which enabled it to adapt to the dynamic environment around the rover and to degrade the rover's mechanical components during the lifetime of its mission.

In [8], we investigated, through the use of goal-oriented navigation experiments, how SODAS builds associations between the sensory modalities to achieve robust autonomous action selection. That work highlighted the fact that intentional behavior is a basic property of intelligence, thus incorporating a cycle of prediction, testing via action, sensing and perceiving, and then assimilating the experienced features. In [9], he brought the ideas together most comprehensively with a look at computational intelligence methods of sensory perception and cognitive functions in animals, humans, and artificial devices. He showed how top-down and bottom-up approaches to modeling the brain can be effective for action selection and conscious decision making in robots and intelligent devices as part of an intentional cognitive cycle.

Conclusion

The prospect of collaborating with Dr. Kozma initially came with the idea of validating my research results, i.e., that signals of computational behavioral dynamics generated by a bioinspired robotics design showed some characteristics similar to those observed in biological brain signal patterns. Our collaboration validated my research results by revealing the applicability of brain-inspired ideas and associated computational models in the context of the

research applications on which I have focused. It also revealed new ways to leverage the complementary aspects of different approaches to build new integrated solutions.

This sparked a collaboration that formed a nexus of behavior-based and brain-based ideas that inspired solutions for autonomous robot control and navigation. Dr. Kozma's theories brought a new scientific approach to our robotics application, inspired us to think about the parallels between artificial behavioral dynamics and neurodynamics, and eventually resulted in the combination of behavioral inspiration and neuro/brain inspiration. In this regard, a lasting impact of his research has been the bringing to light of Freeman's k-sets within the bioinspired robotics community.

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A Cognitive Scientist's View on the Kozma-Freeman Neurodynamics, Bernard Baars, Society of Brain-Mind Sciences, La Jolla, California

I learned of Robert Kozma's and the late Walter J. Freeman's important work through my interest in the psychology and neurobiology of specific conscious experiences, which has now been shown to be dependent on the cortical regions of the brain, primarily the neocortex as well as the hippocampus and rhinal cortex. A wave of modern neuroimaging techniques has clarified a great deal about these broadly "cortical" structures, including their large-scale cellular connectivity in conscious states and the breakdown of connectivity patterns in unconscious states.

Kozma–Freeman Research on Cortical Microstates

The mathematical aspects of cortex are relatively new because high-resolution diffusion tractography has only recently shown fine-level cellular connectivity, i.e., analyzable by network and graph theory. The Kozma–Freeman research presented spectacular results that were quite unexpected to most researchers in the field [1], i.e., cortical “microstates” that seemed to be very widespread and lasted approximately 100 ms, which then collapsed into a brief, chaotic state that allowed for reorganization and a new 100-ms microstate. These phenomena are typically missed when cortical activity is analyzed only by Fourier analysis, although they show up as amplitude-modulation bursts. But, if we view cortex as a spatial analyzer (at least in its dendritic Layer I feltwork), 100-ms microstates and fast collapses emerge, a pattern of results that supports the idea that the cortex operates “near criticality” to be ready for a range of novel inputs while maintaining reasonable stability. This seemed to be a very surprising but robust finding and one that seemed to require advanced mathematics to understand. Dr. Kozma’s work has laid the foundation for that analysis, so I am speaking from an empirical point of view, rather than a strictly mathematical one.

It is widely understood, however, that graph theory and network mathematics now play a crucial role in understanding the “connectome” (the very large and interactive mass of cortical neurons, i.e., the microstructure of what was traditionally called the *white matter*.) Before his passing two years ago, Walter had also focused on nonlinear mathematics to deal with what seemed to be hard-to-analyze properties of cortex. It should be said that we now have both the *structural* connectome (i.e., the street map of cortex) and the *functional* connectome (i.e., the dynamical flow of signal traffic over that street map).

Global Workspace Theory and Consciousness

My own theoretical work is called *global workspace theory* (GWT) [2], [3], and it emerged from multiagent architectures in AI, such as Allan Newell’s work [4]. It was then taken up by many others, notably Stan Franklin, who expanded GWT as a cognitive architecture. GWT predicts the global integration and distribution of brain signals that are perceived as conscious, which are quite different from matched stimuli that are not perceived [5]. GWT therefore seeks some cellular or synaptic mechanism to enable a highly flexible integration among

multiple input streams weighted by prior synaptic learning that is able to recruit large numbers of active cells to propagate a winner-take-all signal. A graph-theoretical or percolation account therefore seems necessary.

In general, these are converging research programs that make graph theory and the cortical connectome look like plausible substrates similar to GWT. Dr. Kozma’s neuropercolation dynamics give a plausible formal account, thus adding to the overall plausibility of these unexpected phenomena. In empirical cognitive neuroscience and biology there are very few major surprises, but the articles by Freeman and Kozma were so different that I had to change my direction. They made a case so robust that conventional (Fourier-analytical) approaches to cortices were missing a major ingredient that emerged very clearly from a simple strip of electrodes on human and animal cortices. This demonstrated an “all-or-none” phase equilibria and collapses near 10 Hz, which is a very significant frequency (and time) range for cortices plausibly linked to alpha and theta waves, now often thought to function as possible carrier waves in cortices for higher-frequency signals. Kozma and Freeman have used the term *cinematic* for this fast-flickering cortical event stream.

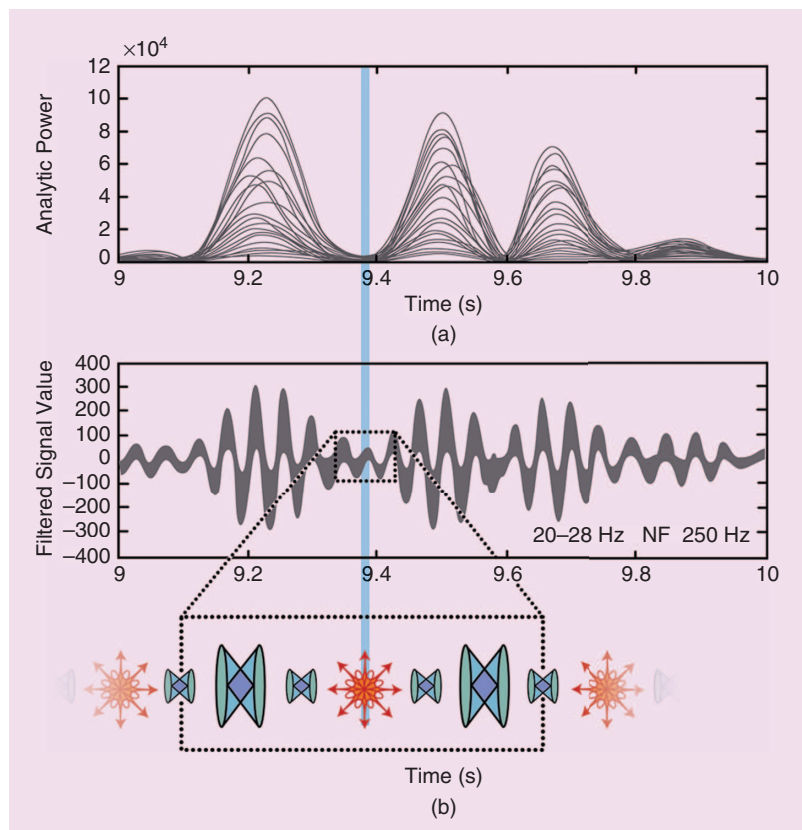


Figure 5. An illustration of the cinematic theory of cognition based on electrocorticogram (ECoG) experiments. (a) The 64 superimposed bandpass-filtered ECoG signals. (b) During the shutter period with desynchronization and singularity (blue bars), phase cones convey the transition from microscopic disorder to macroscopic order [1].

When Layer I of the cortex (called the *dendritic feltwork*) is viewed as a spatial analyzer where Hilbert analysis is appropriate, the “cinematic” character of cortical activity emerges very clearly (see Figure 5). However, the dendrites in Layer I emerge from the top of the pyramidal cell bodies, i.e., the same long-distance neurons that pace the entire cortical system, with roughly 80 billion neurons and trillions of synapses. Since the cinematic all-or-none dynamics emerge in the same cells that also send out long axons to the thalamus and back to the cortex, in the typical fashion of the cortico-thalamic system, it seems likely that a very widespread 10-Hz phase-versus-chaos pattern in Layer I also shapes the resonance of the long axons of the pyramidal neurons. Before his passing, Freeman suggested that the GWT notion of “global distribution and integration” might rely on phase propagation in the cinematic theory. Importantly, I am seeing convergence between the Kozma and Freeman work and GWT, but many questions remain.

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